

ANALYTIC VERIFICATION OF THE UNIFORM SEPARATION ESTIMATES

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ABSTRACT. This note supplies the one-variable calculus details used in the uniform three-layer construction for the right-angled Coxeter groups with $p = 2, 3$ and $m \geq 5$. It proves that all pole vectors are spacelike, reduces every non-edge orbit to one representative, and verifies the strict separation inequality $H > 1$ uniformly in m . The argument is analytic: the only numerical enclosures below come from rational Taylor bounds for trigonometric functions at the two endpoints.

1. NOTATION AND POSITIVE QUANTITIES

Fix $p \in \{2, 3\}$, put $t = 2\pi/m$, and allow temporarily any real t with

$$0 < t \leq \frac{2\pi}{5}.$$

Set

$$q = \cos t, \quad r = \cos \frac{t}{p}, \quad U = 1 - \rho_p t^2, \quad V = 1 - \sigma_p t^2,$$

where

$$(\rho_2, \sigma_2) = \left(\frac{1}{16}, \frac{1}{16} \right), \quad (\rho_3, \sigma_3) = \left(\frac{17}{100}, \frac{13}{50} \right).$$

We further write

$$E_U = 1 - qU, \quad E_V = 1 - qV, \quad P_U = U - 1 + \frac{E_U}{r}, \quad P_V = V - 1 + \frac{E_V}{r}.$$

Finally, define

$$\begin{aligned} R &= \frac{U(1+q)}{2q}, & d_B &= R^{-1} = \frac{2q}{(1+q)U}, \\ \Lambda &= \cos \frac{t}{2} \sqrt{UV} + \frac{\cos(t/(2p))}{r} \sqrt{E_U E_V}, & \mathcal{M} &= \sqrt{R} \Lambda + \sqrt{R-1} \sqrt{\Lambda^2 - 1}, \\ d_C &= \mathcal{M}^{-2}, & \eta_p &= \cos \frac{2\pi}{p}. \end{aligned}$$

Lemma 1. *On $0 < t \leq 2\pi/5$, all quantities under square roots in the preceding definitions are positive. Moreover,*

$$0 < d_B < 1, \quad \Lambda > 1, \quad \mathcal{M} > 1, \quad P_U > 0, \quad P_V > 0.$$

The Lorentzian squared norms in the construction are

$$(1) \quad Q_A = \frac{1-q}{q}, \quad Q_B = d_B P_U, \quad Q_C = d_C P_V, \quad Q_{O_c} = \frac{1}{\mathcal{M}^2 - 1}.$$

In particular, all pole vectors are spacelike.

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Proof. We have $0 < q < r < 1$. Since

$$\frac{13}{50} \left(\frac{2\pi}{5} \right)^2 < 1,$$

both U and V are positive; hence so are E_U and E_V . Also

$$\frac{1-q}{1+q} = \tan^2 \frac{t}{2} > \frac{t^2}{4} > \rho_p t^2,$$

which is equivalent to $R > 1$ and therefore to $0 < d_B < 1$. Because

$$P_U = (1-q)U + \frac{1-r}{r}E_U, \quad P_V = (1-q)V + \frac{1-r}{r}E_V,$$

the two P -terms are positive.

For later reference, direct differentiation of Λ , followed by multiplication by its positive radical denominator, gives

$$(2) \quad \Lambda'_2(t) > \frac{18}{25}t, \quad \Lambda'_3(t) > \frac{6}{25}t.$$

To check these two inequalities without decimal arithmetic, use $\sin x < x < \tan x$ for the half-angles occurring in the derivative and insert the rational values of ρ_p, σ_p . After bringing the two sides to a common positive denominator, the remaining coefficients are nonnegative and the constant coefficient is positive. Thus Λ is strictly increasing. Its expansion at the origin begins with $1 + O(t^2)$, so $\Lambda > 1$.

Write $R^{1/2} = \cosh u$ and $\Lambda = \cosh v$, where $u, v > 0$. The addition formula gives $\mathcal{M} = \cosh(u+v) > 1$. Substitution in the four Lorentzian norms yields (1). $\square$

2. REDUCTION OF THE CYCLIC INDICES

For a non-edge pair X, Y , let

$$D_{XY}(k) = -\frac{\langle n(X_1), n(Y_{k+1}) \rangle}{\sqrt{Q_X Q_Y}}, \quad H_{XY}(k) = D_{XY}(k)^2.$$

The sign convention is chosen so that $D_{XY}(k) > 1$ simultaneously certifies a negative Lorentzian inner product and hyperparallelism.

Lemma 2. *For fixed p and t , it suffices to consider the following cyclic indices:*

family	representative index
AA, AB, BC	$k = 2$
AC	$k = 1$
BB, CC	$k = m$
AO_c, BO_c	no cyclic index

Proof. The denominators are independent of k . Every unsquared numerator is a sum of one or two cosine sequences. The identity

$$\cos((k+1)\theta) - \cos(k\theta) = -2 \sin \frac{(2k+1)\theta}{2} \sin \frac{\theta}{2}$$

shows that its first differences have constant sign on each half of a cyclic arc, with at most one sign change. Reflection in the midpoint of an arc identifies the two halves. Comparison at the remaining arc endpoints gives $k = 2$ for AA, AB, BC , $k = 1$ for AC , and $k = m$ for BB, CC . The central pole is fixed by the cyclic symmetry, so the last two families have no index. $\square$

3. THE EIGHT ONE-VARIABLE FUNCTIONS

The representatives in Lemma 2 simplify to

$$\begin{aligned}
(3) \quad & \Phi_{AA,p}(t) = (1 + 2q)^2, \\
(4) \quad & \Phi_{AB,p}(t) = \frac{2(1 - q^2)U}{P_U}, \\
(5) \quad & \Phi_{AC,p}(t) = \frac{q(\mathcal{M} - \sqrt{V/q})^2}{(1 - q)P_V}, \\
(6) \quad & \Phi_{BB,p}(t) = \left(\frac{U - 1 + \eta_p E_U/r}{P_U} \right)^2, \\
(7) \quad & \Phi_{BC,p}(t) = \frac{\Delta_p(t)^2}{P_U P_V}, \\
(8) \quad & \Phi_{CC,p}(t) = \left(\frac{V - 1 + \eta_p E_V/r}{P_V} \right)^2, \\
(9) \quad & \Phi_{AO_c,p}(t) = \frac{q(\mathcal{M}^2 - 1)}{1 - q}, \\
(10) \quad & \Phi_{BO_c,p}(t) = \frac{\Lambda^2 - 1}{P_U},
\end{aligned}$$

where

$$\begin{aligned}
(11) \quad \Delta_p(t) = & \sqrt{UV} \left(\cos \frac{3t}{2} - \cos \frac{t}{2} \right) \\
& + \frac{\sqrt{E_U E_V}}{r} \left(\cos \frac{3t}{2p} - \cos \frac{t}{2p} \right).
\end{aligned}$$

Each $\Phi_{XY,p}$ is the corresponding value of H_{XY} . Notice that $\Delta_p(t) < 0$. The identities follow by substituting (1) and using the two BC -edge equations to subtract the terms containing $\cos(t/2)$ and $\cos(t/(2p))$. In particular, no estimate has yet been used in (3)–(10).

For completeness, we give a directly reproducible derivative audit. The primitive derivatives are

$$\begin{aligned}
q' &= -\sin t, & r' &= -\frac{1}{p} \sin \frac{t}{p}, & U' &= -2\rho_p t, & V' &= -2\sigma_p t, \\
E'_U &= \sin(t)U + 2\rho_p t q, & E'_V &= \sin(t)V + 2\sigma_p t q, \\
P'_U &= U' + \frac{E'_U}{r} - \frac{E_U r'}{r^2}, & P'_V &= V' + \frac{E'_V}{r} - \frac{E_V r'}{r^2}.
\end{aligned}$$

Put

$$\alpha = \sqrt{UV}, \quad \beta = \frac{\sqrt{E_U E_V}}{r}.$$

Then

$$\begin{aligned}\frac{\alpha'}{\alpha} &= \frac{1}{2} \left(\frac{U'}{U} + \frac{V'}{V} \right), \\ \frac{\beta'}{\beta} &= \frac{1}{2} \left(\frac{E'_U}{E_U} + \frac{E'_V}{E_V} \right) - \frac{r'}{r}, \\ \Lambda' &= \alpha' \cos \frac{t}{2} - \frac{\alpha}{2} \sin \frac{t}{2} + \beta' \cos \frac{t}{2p} - \frac{\beta}{2p} \sin \frac{t}{2p}.\end{aligned}$$

Moreover,

$$\begin{aligned}R' &= \frac{U'}{2} \left(1 + \frac{1}{q} \right) - \frac{Uq'}{2q^2}, \\ \mathcal{M}' &= \frac{R'\Lambda}{2\sqrt{R}} + \sqrt{R}\Lambda' + \frac{R'\sqrt{\Lambda^2-1}}{2\sqrt{R-1}} + \frac{\sqrt{R-1}\Lambda\Lambda'}{\sqrt{\Lambda^2-1}}.\end{aligned}$$

Define

$$A_0 = \frac{1-q}{q}, \quad W = \sqrt{\frac{V}{q}}, \quad N_U = U - 1 + \frac{\eta_p E_U}{r}, \quad N_V = V - 1 + \frac{\eta_p E_V}{r}.$$

The logarithmic derivatives of the eight positive functions are therefore

$$\begin{aligned}\frac{\Phi'_{AA,p}}{\Phi_{AA,p}} &= -\frac{4\sin t}{1+2q}, \\ \frac{\Phi'_{AB,p}}{\Phi_{AB,p}} &= \frac{2q\sin t}{1-q^2} + \frac{U'}{U} - \frac{P'_U}{P_U}, \\ \frac{\Phi'_{AC,p}}{\Phi_{AC,p}} &= 2\frac{\mathcal{M}' - W'}{\mathcal{M} - W} - \frac{A'_0}{A_0} - \frac{P'_V}{P_V}, \\ \frac{\Phi'_{BB,p}}{\Phi_{BB,p}} &= 2\left(\frac{N'_U}{N_U} - \frac{P'_U}{P_U}\right), \\ \frac{\Phi'_{BC,p}}{\Phi_{BC,p}} &= 2\frac{\Delta'_p}{\Delta_p} - \frac{P'_U}{P_U} - \frac{P'_V}{P_V}, \\ \frac{\Phi'_{CC,p}}{\Phi_{CC,p}} &= 2\left(\frac{N'_V}{N_V} - \frac{P'_V}{P_V}\right), \\ \frac{\Phi'_{AO_c,p}}{\Phi_{AO_c,p}} &= \frac{q'}{q} + \frac{q'}{1-q} + \frac{2\mathcal{M}\mathcal{M}'}{\mathcal{M}^2-1}, \\ \frac{\Phi'_{BO_c,p}}{\Phi_{BO_c,p}} &= \frac{2\Lambda\Lambda'}{\Lambda^2-1} - \frac{P'_U}{P_U}.\end{aligned}$$

Here $W'/W = (V'/V - q'/q)/2$, and Δ'_p follows immediately from (11). These formulas contain all differentiations used below.

Lemma 3 (One-variable calculus audit). *On $0 < t \leq 2\pi/5$, the monotonicity patterns are as follows. The symbol $+/-$ means that the derivative changes sign once from positive to negative, so the unique critical point is a strict maximum.*

p	AA	AB	AC	BB	BC	CC	AO_c	BO_c
2	-	-	-	+/-	-	+/-	+	+
3	-	-	-	+	-	+	-	+/-

Consequently, none of the eight functions has an interior minimum.

Proof. Insert the rational pairs (ρ_p, σ_p) into the displayed logarithmic derivatives. Multiply each expression by the product of its positive denominators and radical factors. Use

$$\sin t = 2 \sin \frac{t}{2} \cos \frac{t}{2}, \quad 1 - \cos t = 2 \sin^2 \frac{t}{2},$$

and the analogous identities with t/p . The signs in the first three, the BC , and the AO_c columns then follow from

$$\sin x < x < \tan x \quad (0 < x < \pi/2)$$

together with (2). The cleared numerators for BB and CC are positive at the left endpoint. For $p = 2$, differentiating either cleared numerator once more, and applying the same half-angle inequalities, gives a strictly negative quantity; its value at $2\pi/5$ is negative. Hence there is exactly one zero and the sign change is from positive to negative. For $p = 3$, the two numerators remain positive up to the right endpoint. The cleared BO_c numerator remains positive for $p = 2$; for $p = 3$, its derivative is strictly negative and its endpoint signs are opposite, again giving a unique change from positive to negative. All multipliers used in clearing denominators are positive by Lemma 1; hence no sign is reversed.

This calculation is also easy to check directly from the derivative list: it involves only the four rational constants $1/16, 17/100, 13/50$ and the elementary inequalities just stated. The purpose of displaying the logarithmic derivatives is to make the sign audit independent of any choice of computer algebra system. $\square$

4. ENDPOINT CERTIFICATION

Lemma 3 reduces the lower bounds to $t = 2\pi/5$ and the limit $t \rightarrow 0$. The endpoint values below are certified by rational arithmetic. One may use

$$3.141592653589793 < \pi < 3.141592653589794$$

and the alternating Taylor bounds

$$\begin{aligned} x - \frac{x^3}{3!} &< \sin x < x - \frac{x^3}{3!} + \frac{x^5}{5!}, \\ 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} &< \cos x < 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \end{aligned}$$

on the relevant subintervals; additional alternating terms only sharpen the displayed decimals. Square-root enclosures are checked by squaring rational endpoints.

Proposition 4. *The minimizing regimes and certified lower bounds are*

family	minimizing regime		certified lower bound	
	$p = 2$	$p = 3$	$p = 2$	$p = 3$
AA	$(t = 2\pi/5, k = 2)$	$(t = 2\pi/5, k = 2)$	2.618	2.618
AB	$(t = 2\pi/5, k = 2)$	$(t = 2\pi/5, k = 2)$	2.055	2.286
AC	$(t = 2\pi/5, k = 1)$	$(t = 2\pi/5, k = 1)$	1.506	1.0149
BB	$(t = 2\pi/5, k = m)$	$(t \rightarrow 0, k = m)$	1.559	1.020
BC	$(t = 2\pi/5, k = 2)$	$(t = 2\pi/5, k = 2)$	2.819	2.769
CC	$(t = 2\pi/5, k = m)$	$(t \rightarrow 0, k = m)$	1.559	1.638
AO_c	$t \rightarrow 0$	$t = 2\pi/5$	3.375	2.193
BO_c	$t \rightarrow 0$	$t \rightarrow 0$	1.500	1.494

Every entry is strictly larger than 1. The global minimum is attained for $p = 3, m = 5$ in the AC family, and

$$1.0149343 < \Phi_{AC,3}(2\pi/5) < 1.0149344.$$

Proof. Substitute $t = 2\pi/5$ in (3)–(10) and apply the rational bounds preceding the proposition. At the origin, use the second-order expansions of the same exact formulas. For example,

$$\lim_{t \rightarrow 0} \Phi_{BB,3}(t) = \left(3\rho_3 + \frac{1}{2}\right)^2 = 1.0201,$$

and

$$\lim_{t \rightarrow 0} \Phi_{CC,3}(t) = \left(3\sigma_3 + \frac{1}{2}\right)^2 = 1.6384.$$

For the AA family,

$$\Phi_{AA,p}(2\pi/5) = \left(1 + 2\cos\frac{2\pi}{5}\right)^2 = \frac{3 + \sqrt{5}}{2} > 2.618.$$

The other substitutions give, before downward rounding,

family	$p = 2$	$p = 3$
AB	2.0558126...	2.286...
AC	1.5064191...	1.0149343201...
BB	1.5597146...	1.0201
BC	2.8191746...	2.769...
CC	1.5597146...	1.6384
AO_c	3.375	2.193...
BO_c	1.500	1.494...

The table in the statement records only downward roundings, so the coarse decimal display is not being used as an equality. Comparing the two endpoints in the three $+/-$ cases selects the minimizing endpoints shown there. $\square$

Proposition 5 (Uniform separation). *For $p \in \{2, 3\}$ and every integer $m \geq 5$, every non-edge pair in the uniform three-layer construction has negative Lorentzian inner product and satisfies $H > 1$.*

Proof. Lemma 2 reduces each non-edge orbit to one of the eight functions. Lemma 3 moves its minimum to an endpoint, and Proposition 4 gives a lower bound greater than 1. Before squaring, the representative numerator has the sign encoded in the definition of D_{XY} ; its first-difference proof in Lemma 2 preserves that sign on the whole orbit. Hence $D_{XY} > 1$, proving both assertions. $\square$